

Q3d: Expenditure Minimization with Log Utility

The Question

Given: Melsi has preferences $u(C_1, C_2) = v(C_1) + v(C_2)$ where v is increasing and concave. For part (d), we specialize to $v(C) = \ln C$.

Question: What bundle (C_1, C_2) most efficiently (i.e., cheaply) attains utility u^* ?

What This Question Is Really Asking

This is an **expenditure minimization problem**, NOT a utility maximization problem!

Standard problem: Given prices and income, maximize utility

$$\max_{C_1, C_2} u(C_1, C_2) \quad \text{subject to} \quad p_1 C_1 + p_2 C_2 = m$$

This problem (dual): Given prices and a target utility level u^* , minimize expenditure

$$\min_{C_1, C_2} p_1 C_1 + p_2 C_2 \quad \text{subject to} \quad u(C_1, C_2) = u^*$$

Key difference:

- Standard: "Here's your budget, get the most utility"
- Expenditure minimization: "Here's the utility you need to reach, spend as little as possible"

Step-by-Step Solution

Step 1: Recall the setup from part (c)

From the budget constraint in standard form: $2C_1 + C_2 \leq 2w_1$

So the prices are:

- $p_1 = 2$ (price of C_1)
- $p_2 = 1$ (price of C_2)

With $v(C) = \ln C$, the utility function is:

$$u(C_1, C_2) = \ln C_1 + \ln C_2$$

Step 2: Set up the expenditure minimization problem

We want to:

$$\min_{C_1, C_2} 2C_1 + C_2$$

subject to:

$$\ln C_1 + \ln C_2 = u^*$$

Step 3: Find the optimal condition (MRS = price ratio)

At the optimal bundle, the marginal rate of substitution must equal the price ratio:

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

Calculate marginal utilities:

$$MU_1 = \frac{\partial u}{\partial C_1} = \frac{\partial}{\partial C_1}(\ln C_1 + \ln C_2) = \frac{1}{C_1}$$

$$MU_2 = \frac{\partial u}{\partial C_2} = \frac{\partial}{\partial C_2}(\ln C_1 + \ln C_2) = \frac{1}{C_2}$$

So the optimal condition is:

$$\frac{MU_1}{MU_2} = \frac{1/C_1}{1/C_2} = \frac{C_2}{C_1} = \frac{p_1}{p_2} = \frac{2}{1} = 2$$

First condition:

$$\frac{C_2}{C_1} = 2 \quad \Rightarrow \quad \boxed{C_2 = 2C_1}$$

Step 4: Use the utility constraint

The second condition is that we must achieve utility u^* :

$$\ln C_1 + \ln C_2 = u^*$$

Step 5: Substitute $C_2 = 2C_1$ into the utility constraint

$$\begin{aligned}\ln C_1 + \ln(2C_1) &= u^* \\ \ln C_1 + \ln 2 + \ln C_1 &= u^* \quad (\text{using } \ln(ab) = \ln a + \ln b) \\ 2 \ln C_1 + \ln 2 &= u^* \\ \ln(C_1^2) + \ln 2 &= u^* \quad (\text{using } n \ln a = \ln(a^n)) \\ \ln(2C_1^2) &= u^* \quad (\text{using } \ln a + \ln b = \ln(ab))\end{aligned}$$

Step 6: Solve for C_1

Exponentiate both sides:

$$2C_1^2 = e^{u^*}$$

Divide by 2:

$$C_1^2 = \frac{e^{u^*}}{2} = \frac{1}{2}e^{u^*}$$

Take the square root:

$$C_1 = \sqrt{\frac{1}{2}e^{u^*}}$$

We can rewrite this using exponent rules: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{e^{u^*}} = e^{u^*/2}$

Solution for C_1 :

$$C_1 = \sqrt{\frac{1}{2}e^{u^*/2}} = \frac{1}{\sqrt{2}}e^{u^*/2} = \boxed{\sqrt{\frac{1}{2}e^{u^*/2}}}$$

Or equivalently:

$$\boxed{C_1 = \frac{e^{u^*/2}}{\sqrt{2}}}$$

Step 7: Find C_2

Using $C_2 = 2C_1$:

$$\begin{aligned}C_2 &= 2C_1 \\&= 2 \cdot \sqrt{\frac{1}{2}e^{u^*/2}} \\&= 2 \cdot \frac{1}{\sqrt{2}} \cdot e^{u^*/2} \\&= \frac{2}{\sqrt{2}} \cdot e^{u^*/2} \\&= \sqrt{2} \cdot e^{u^*/2} \quad (\text{since } \frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2})\end{aligned}$$

We can also write this as:

$$C_2 = \sqrt{2 \cdot e^{u^*/2}}$$

Solution for C_2 :

$$C_2 = \sqrt{2}e^{u^*/2} = \sqrt{2e^{u^*/2}}$$

Final Answer

The bundle that most cheaply attains utility u^* is:

$$\begin{aligned}C_1 &= \sqrt{\frac{1}{2}e^{u^*/2}} = \frac{e^{u^*/2}}{\sqrt{2}} \\C_2 &= \sqrt{2e^{u^*/2}} = \sqrt{2}e^{u^*/2}\end{aligned}$$

Check the ratio:

$$\frac{C_2}{C_1} = \frac{\sqrt{2}e^{u^*/2}}{\frac{e^{u^*/2}}{\sqrt{2}}} = \sqrt{2} \cdot \sqrt{2} = 2 \quad \checkmark$$

Check the utility:

$$\begin{aligned}\ln C_1 + \ln C_2 &= \ln \left(\frac{e^{u^*/2}}{\sqrt{2}} \right) + \ln \left(\sqrt{2}e^{u^*/2} \right) \\&= \left(\frac{u^*}{2} - \frac{1}{2} \ln 2 \right) + \left(\frac{1}{2} \ln 2 + \frac{u^*}{2} \right) \\&= \frac{u^*}{2} + \frac{u^*}{2} \\&= u^* \quad \checkmark\end{aligned}$$

Why This Makes Economic Sense

The Optimal Ratio

The condition $C_2 = 2C_1$ tells us:

- Good 1 costs twice as much as good 2 ($p_1 = 2, p_2 = 1$)
- To minimize cost while staying on the same indifference curve, we should consume twice as much of the cheaper good
- This is because with log utility, the MRS equals the consumption ratio: $\frac{C_2}{C_1} = \frac{p_1}{p_2}$

Connection to Standard Problem

In the standard utility maximization problem:

- We're given income m and prices (p_1, p_2)
- We maximize $u(C_1, C_2)$ subject to $p_1C_1 + p_2C_2 = m$
- We find the optimal bundle (C_1^*, C_2^*) that achieves some utility level u^*

In the expenditure minimization problem (this question):

- We're given the target utility u^* and prices (p_1, p_2)
- We minimize $p_1C_1 + p_2C_2$ subject to $u(C_1, C_2) = u^*$
- We find the cheapest bundle that achieves u^*

Key insight: These two problems give the *same* bundle! The optimal bundle from utility maximization with income m is also the cheapest way to achieve the resulting utility level.

Alternative Approach: Lagrangian

You can also solve this using the Lagrangian method:

$$\mathcal{L} = 2C_1 + C_2 - \lambda(\ln C_1 + \ln C_2 - u^*)$$

First-order conditions:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial C_1} &= 2 - \frac{\lambda}{C_1} = 0 \quad \Rightarrow \quad \lambda = 2C_1 \\ \frac{\partial \mathcal{L}}{\partial C_2} &= 1 - \frac{\lambda}{C_2} = 0 \quad \Rightarrow \quad \lambda = C_2 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= \ln C_1 + \ln C_2 - u^* = 0\end{aligned}$$

From the first two conditions: $2C_1 = C_2$, which matches what we found! Then substitute into the constraint and solve as before.